MicroSense AI-Cam Report

AI-assisted optical monitoring report for preliminary screening of microplastic-like particle candidates in water samples.

1. Sample Summary

Report Generated	15 May 2026, 10:11 AM
Sample ID	1
Sample Source	Lake water
Chamber Volume	50 ml
File Type	image
Monitoring Risk Level	**Moderate**

2. Detection and Monitoring Results

Detected Candidates	12
Estimated Particles / Litre	240
MSMI Score	31
MPI Score	31
Confidence Score	75
Particle Size Category	100–500 µm estimated (prototype)
Average Particle Area	1434.08
Average Brightness	48.39

3. Hybrid Validation Results

Raw Detection Count	12
Accepted Candidate Count	12
Rejected Candidate Count	0
Hybrid Validation Score	71.06
Filter Summary	All 12 detected particle candidates passed the hybrid validation filter.

4. Image Quality Assessment

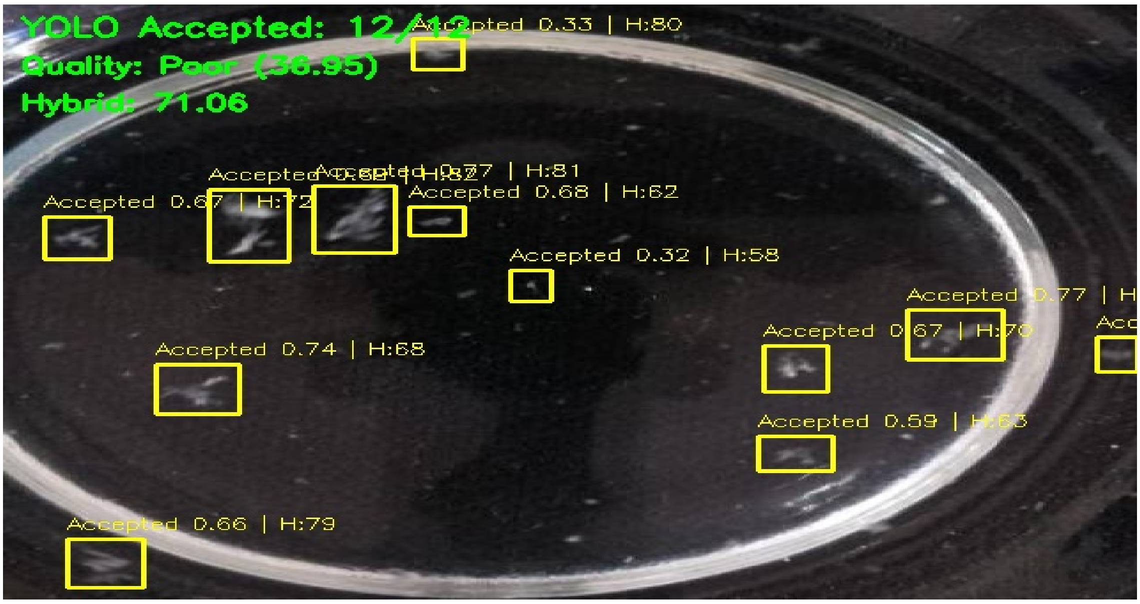
Image Quality Status	Poor
Image Quality Score	36.95
Focus Score	28.42
Brightness Score	32.37
Contrast Score	49.22
Overexposed Percent	0.28
Underexposed Percent	39.47
Quality Warning	Image may be blurry. Image is too dark. High underexposure detected.

5. Risk Interpretation

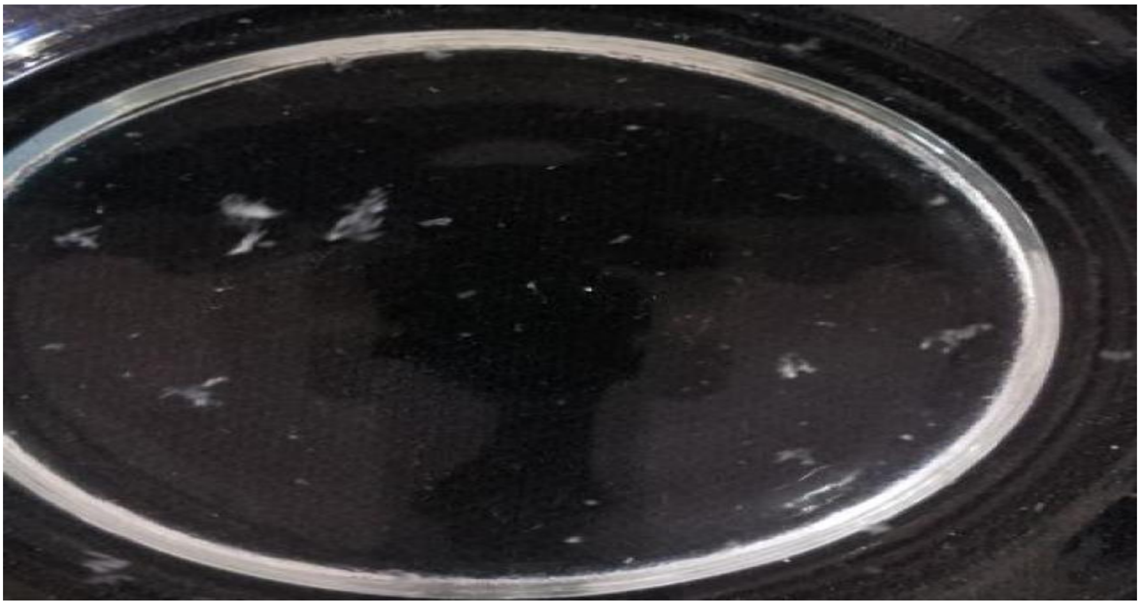
Source Risk Factor	1.25
Concentration Score	75
Size Score	40
Concentration-Only Risk Level	High
Risk Explanation	MSMI is calculated using estimated particle concentration, detected count, average partic
Recommendation	Moderate monitoring concern. Retest the sample and check possible contamination sourc
Notes	—

6. Image Evidence

Processed Output Image



Original Uploaded Image



7. Scientific Scope and Disclaimer

This report is generated by MicroSense AI-Cam, a prototype-level AI-assisted optical monitoring system. The reported objects represent microplastic-like particle candidates identified from image-based visual features. The system does not chemically confirm polymer composition and does not replace certified laboratory techniques such as FTIR spectroscopy, Raman spectroscopy, or advanced microscopic analysis. The generated values, including estimated concentration, MSMI score, confidence score, and monitoring risk level, should be interpreted only as preliminary screening indicators and not as regulatory-grade or laboratory-certified measurements.

8. System Pipeline

Water sample image acquisition → YOLO26n-based candidate detection → OpenCV-based fallback support where required → hybrid visual validation → image quality assessment → monitoring score calculation → database storage → dashboard and report generation.